ACF · Cost–volume–profit and operating decisions — Question Patterns and Methods

Based on 20 past-exam parts, 6 years (2020–2025), their marking keys and the examiners' reports.

First: where marks are lost most

- Unclear or missing workings. The examiners' reports say many candidates did not show clear calculations, and some treated an intermediate working as the final answer and did not write the calculated amount in the answer space. Keys award marks for each intermediate amount (unit contribution, sales mix, weighted component, break-even units), so these marks are lost even when the method is right (Patterns 1–4, 8).
- Including or excluding the wrong costs. Keys award separate marks for fixed costs that are relevant, additional fixed costs, extra per-unit variable costs and opportunity costs, and students often miss one of these (Patterns 4, 6, 7).
- Using the wrong base or denominator for a mix or allocation. Keys reward the specified base, such as total bottles or total kilometres, and a different base loses the mix marks and every amount that follows (Pattern 2).
- Not applying a change to the figures the question sets. In closure and make-or-buy decisions, keys reward the reduced or retained fixed costs after the change, and students who keep the original figures lose those marks (Patterns 7, 8).
- Explanations that do not use the stimulus. The 2024 examiners' report says extended answers lacked the required detail, did not refer to the information provided, and did not use accounting and finance terminology (Pattern 9).

Pattern 1: Calculate total, unit contribution and contribution ratio

How often: 4/6 papers · 2.0 marks when it appears · 0.6% of a paper.

Recognise it: a stem gives a selling price (or total sales revenue), a list of costs and a number of units, and asks for the contribution margin, the contribution per unit or the contribution ratio. Costs are often mixed with fixed costs and profit, so sort them first. Multiple-choice versions give the profit figure as a distractor trap.

Method

1. Find the selling price per unit, or divide total sales revenue by the units sold.
2. List only the variable costs (for example labour, materials and energy that change with output). Leave out fixed costs and profit.
3. Unit contribution = selling price − variable cost per unit. For a total, contribution = sales revenue − total variable costs.
4. Contribution ratio = total contribution ÷ sales revenue × 100, rounded as the question requires.
5. Write each intermediate amount on its own line, then the final answer.

What the key rewards: for a multi-product part, the selling price and the unit cost for each product, then the contribution for each. For a two-mark part, the selling price, then the deduction of each variable cost. For multiple-choice items, the single correct option.

Common errors: treating a fixed cost as a variable cost; subtracting profit; giving the contribution ratio without the workings; rounding before the calculation is finished.

Past questions: 2023 SA Q18(c), 2022 MC Q7, 2021 SA Q18(a), 2020 MC Q6

Pattern 2: Calculate product or activity mix

How often: 2/6 papers · 11.0 marks when it appears · 0.9% of a paper.

Recognise it: the stem asks for a sales mix, a share of total sales or activity, or an allocation of a shared cost by a base such as kilometres travelled. It may then ask you to apply that mix to a required total, such as the break-even volume.

Method

1. Identify the total of the specified base. Add the components if the total is not given (for example total bottles sold across all products).
2. For each product or region, mix = its amount ÷ the total, shown as a percentage.
3. To allocate a shared cost, multiply the cost by each mix percentage.
4. To apply a mix to a required total, multiply the total by each product's mix percentage.
5. Carry the rounded mix percentages into the later steps that depend on them.

What the key rewards: for the sixteen-mark regional part, the fixed-cost allocation to each region, each region's mix percentage, and the total kilometres used as the denominator. For the four-mark part, the total volume, then the mix for each product. For the two-mark part, the break-even total and the mix applied to it.

Common errors: using a base other than the one the question specifies; allocating a shared cost without multiplying by the mix; leaving out the total when showing the mix; applying a mix to the wrong total.

Past questions: 2024 SA Q20(a), 2021 SA Q18(b), 2021 SA Q18(e)

Pattern 3: Calculate weighted average contribution

How often: 1/6 papers · 4.0 marks when it appears · 0.3% of a paper.

Recognise it: the stem gives a contribution margin for each product and the sales mix, and asks for one average contribution for the whole range.

Method

1. Confirm the sales mix percentage for each product.
2. Multiply each product's contribution margin by its mix to get a weighted component.
3. Add the weighted components. The sum is the weighted average contribution per composite unit.
4. Round the calculations as the question directs, and show each component.

What the key rewards: one mark for each product's weighted component, and one for the final weighted average.

Common errors: taking a simple average of the contributions instead of a weighted one; applying the mix to the wrong product; giving only the final average without the components.

Past questions: 2021 SA Q18(c)

Pattern 4: Calculate break-even or target-profit sales

How often: 3/6 papers · 1.7 marks when it appears · 0.4% of a paper.

Recognise it: the stem asks for a break-even point in units, bottles or revenue, or for the sales needed to reach a target profit. Multiple-choice items may test the definition of break-even.

Method

1. Identify the fixed costs, including any additional fixed costs that the question introduces.
2. Identify the contribution per unit. For a product range, use the weighted average contribution from Pattern 3.
3. Break-even units = total fixed costs ÷ contribution per unit. For a target profit, add the target to the fixed costs before dividing.
4. If revenue is asked for, multiply the units by the selling price.
5. State the answer in the unit the question asks for.

What the key rewards: the base fixed costs, the additional fixed costs, the weighted contribution margin, and the break-even answer in bottles. For multiple-choice items, the definition of break-even: the level at which

sales revenue equals total costs, and the number of units at which contribution covers fixed costs.

Common errors: leaving out additional fixed costs; using an unweighted contribution for a product mix; giving revenue when units are asked for, or units when revenue is asked for.

Past questions: 2025 MC Q12, 2021 SA Q18(d), 2020 MC Q15

Pattern 5: Calculate and explain margin of safety

How often: 2/6 papers · 1.5 marks when it appears · 0.2% of a paper.

Recognise it: the stem gives budgeted sales and asks for the margin of safety in units, or asks you to describe what the measure shows.

Method

1. Find the break-even units = fixed costs ÷ contribution per unit.
2. Margin of safety (units) = budgeted sales − break-even units.
3. If a percentage is asked, margin of safety ÷ budgeted sales × 100.
4. For a written part, say the margin of safety measures how far sales exceed the break-even point, in units or as a percentage.

What the key rewards: for the calculation, the correct margin in units. For the description, the idea that the margin is the level of sales above break-even.

Common errors: giving the break-even point instead of the margin; subtracting in the wrong direction; describing the margin without linking it to the break-even point.

Past questions: 2025 MC Q2, 2022 SA Q16(c)

Pattern 6: Evaluate a special order

How often: 1/6 papers · 11.0 marks when it appears · 0.8% of a paper.

Recognise it: a customer offers a one-off order at a price different from the regular price, often with a quantity and some new costs. The question asks for the gain or loss and whether to accept or reject.

Method

1. Special-order revenue = special price × quantity.
2. Relevant variable cost = unit variable cost plus any extra variable cost for the order, multiplied by quantity.
3. Contribution before other costs = special-order revenue − relevant variable costs.
4. Deduct new fixed costs that arise only because of the order.
5. Deduct the opportunity cost, which is the contribution lost from regular sales displaced by the order (units displaced × regular contribution per unit), if capacity is limited.
6. The result is the gain (positive) or loss (negative). Accept if the gain is positive, and state the recommendation.

What the key rewards (eleven marks, in order): the special selling price; the order quantity; the unit variable cost; the extra variable cost per unit; total special-order variable cost; the regular selling price used for the opportunity cost; the regular variable cost used for the opportunity cost; the contribution before extra fixed and opportunity costs; the new fixed costs deducted; the opportunity cost deducted to give the gain; and the recommendation.

Common errors: leaving out the extra per-unit variable cost; using the regular price in the special-order calculation; omitting the opportunity cost; including existing fixed costs that do not change; not stating a recommendation.

Past questions: 2022 SA Q16(a)

Pattern 7: Evaluate make-or-buy alternatives

How often: 1/6 papers · 9.0 marks when it appears · 0.7% of a paper.

Recognise it: the business considers buying a component from a supplier instead of making it, and the stem gives the supplier price, the cost of making each unit, and a change in fixed costs.

Method

1. Make cost = variable cost per unit × units + fixed costs that continue if the item is made.
2. Buy cost = supplier price × units + fixed costs that continue if the item is bought, after any reduction.
3. Compare the two totals. The option with the lower cost gives more profit.
4. Show the difference and recommend the option.

What the key rewards (nine marks): the variable cost per component and the number of components for the make option; the fixed costs in the make option; the supplier price and the number of components for the buy option; the allocated fixed costs before the reduction; the reduction applied to the buy option; the difference in profit; and the recommendation.

Common errors: applying the fixed-cost reduction to the make option, or leaving it out of the buy option; keeping allocated fixed costs that do not change; comparing per-unit costs only; not stating the difference in profit.

Past questions: 2022 SA Q16(b)

Pattern 8: Evaluate segment profitability and closure

How often: 1/6 papers · 27.0 marks when it appears · 1.4% of a paper.

Recognise it: the stem gives revenue and costs for each region or service, asks for profit or loss by segment, then asks for the effect on total profit if one segment closes and whether to recommend it.

Method

1. For each segment, profit or loss = revenue − labour − materials − allocated fixed costs.
2. Total profit = the sum of the segment results.
3. If a segment is closed, remove its revenue and its direct costs from the totals.
4. Recalculate the fixed costs that remain. The key treats part of the original fixed costs as the share for the retained services.
5. New total profit − original total profit = the effect of closure.
6. Recommend closing or keeping the segment, and justify it with the change in profit or contribution.

What the key rewards: for the sixteen-mark part, the revenue, labour and materials for each region. For the nine-mark part: the profit of each retained region, the overall profit, the remaining revenue, labour and materials after closure, the share of fixed costs retained, the retained fixed costs, and the increase in profit. For the two-mark part, a recommendation with a justification that refers to profit or contribution.

Common errors: keeping the original fixed costs unchanged after closure; treating the segment's loss as the full effect on profit; leaving out the remaining revenue or direct costs; recommending without a justification.

Past questions: 2024 SA Q20(a), 2024 SA Q20(b), 2024 SA Q20(c)

Pattern 9: Explain cost-volume-profit changes

How often: 3/6 papers · 4.0 marks when it appears · 0.6% of a paper.

Recognise it: the stem changes sales volume, price, a variable cost or fixed costs and asks for the effect on profit or break-even. Extended questions give a stimulus article and ask you to describe cost behaviour and explain the effect.

Method

1. Identify which figures change and which stay the same.
2. For a direction question, decide the effect on profit: change in profit = change in contribution − change in fixed costs.
3. For a written part, describe how variable costs change in total with activity (and per unit they stay the same), and how fixed costs stay the same in total within the relevant range.

4. Link the change to break-even: break-even = fixed costs ÷ contribution per unit. State the effect with figures from the stimulus.
5. Give a clear conclusion about the direction of the effect on profit or break-even.

What the key rewards: for the seven-mark part, an explanation of how rising variable and fixed costs raise the break-even sales point, with the new break-even sales figure. For the four-mark part, an explanation of how the link between activity, costs and profit affects performance, illustrated with an example from the article. Multiple-choice items reward the correct direction after analysing each distractor.

Common errors: describing cost behaviour without linking it to break-even or profit; not referring to the stimulus; using vague terms instead of accounting terminology; leaving out detail, as the 2024 examiners' report noted for extended answers.

Past questions: 2024 MC Q9, 2023 EA Q21(a), 2020 EA Q21(a)

Pattern 10: Solve a cost-volume-profit relationship

How often: no past-exam parts in the evidence pack · no mean marks · no share of a paper.

Recognise it: the stem gives most of the profit relationship (sales price, variable cost, fixed costs, profit or units) and asks you to solve for one unknown under the stated conditions.

Method

1. Write the profit relationship: Profit = (selling price − variable cost per unit) × units sold − fixed costs.
2. Substitute the known figures.
3. Rearrange to isolate the unknown. For example, selling price = (profit + fixed costs) ÷ units + variable cost per unit.
4. Solve, and check the answer by substituting it back into the relationship.

What the key rewards: no marking key is in the evidence pack, so the awards for this pattern are not recorded.

Common errors: none recorded in the evidence pack.

Past questions: none in the evidence pack.